Penguin Data Exploration & Visualization
DS 201 – Tools for Data Science
Final Project Report

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Date: March 6, 2026

1. Introduction (About the Dataset)

The dataset used in this project is the Palmer Penguins dataset, which contains biological measurements of penguins from three different species: Adelie, Chinstrap, and Gentoo. The data includes several variables such as island location, bill length, bill depth, flipper length, body mass, and sex.

The purpose of this analysis is to explore patterns and relationships within the dataset using exploratory data analysis techniques. By analyzing these measurements, we can better understand how physical characteristics differ across penguin species and locations.

Python was used for the analysis, with the pandas library used for data exploration and cleaning, and Matplotlib used to create visualizations that help reveal patterns within the data.

2. Data Exploration

The dataset was first explored to understand its structure and contents. It contains multiple columns including species, island, bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g, and sex.

Initial exploration involved reviewing the dataset preview and identifying the types of variables included. Numerical variables such as bill length, bill depth, flipper length, and body mass represent physical measurements of penguins. Categorical variables such as species, island, and sex describe classifications within the dataset.

Exploring the dataset helps identify potential patterns, differences between species, and any missing values that may require cleaning before performing further analysis.

3. Data Cleaning

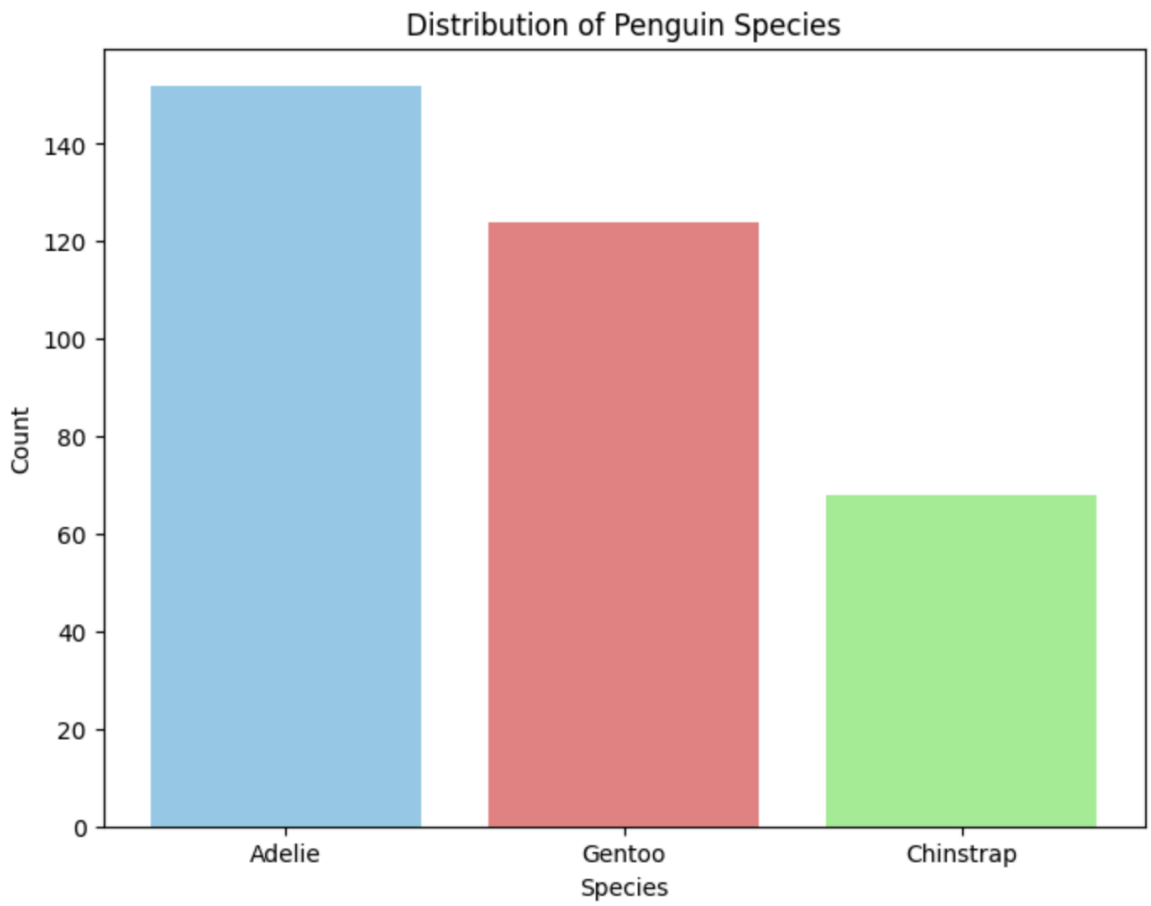
During the data cleaning stage, the dataset was reviewed to identify missing or incomplete values. Some rows contained missing measurements for certain variables.

To ensure the dataset remained usable for analysis, missing values in numerical columns were replaced using the median value of the column. This method helps maintain the general distribution of the data while preventing extreme values from influencing the results.

For categorical variables such as species or sex, missing values were replaced using the mode, which represents the most frequent category in the dataset. After cleaning the data, the dataset was ready for visualization and further analysis.

4. Data Visualization

Figure 1 – Distribution of Penguin Species

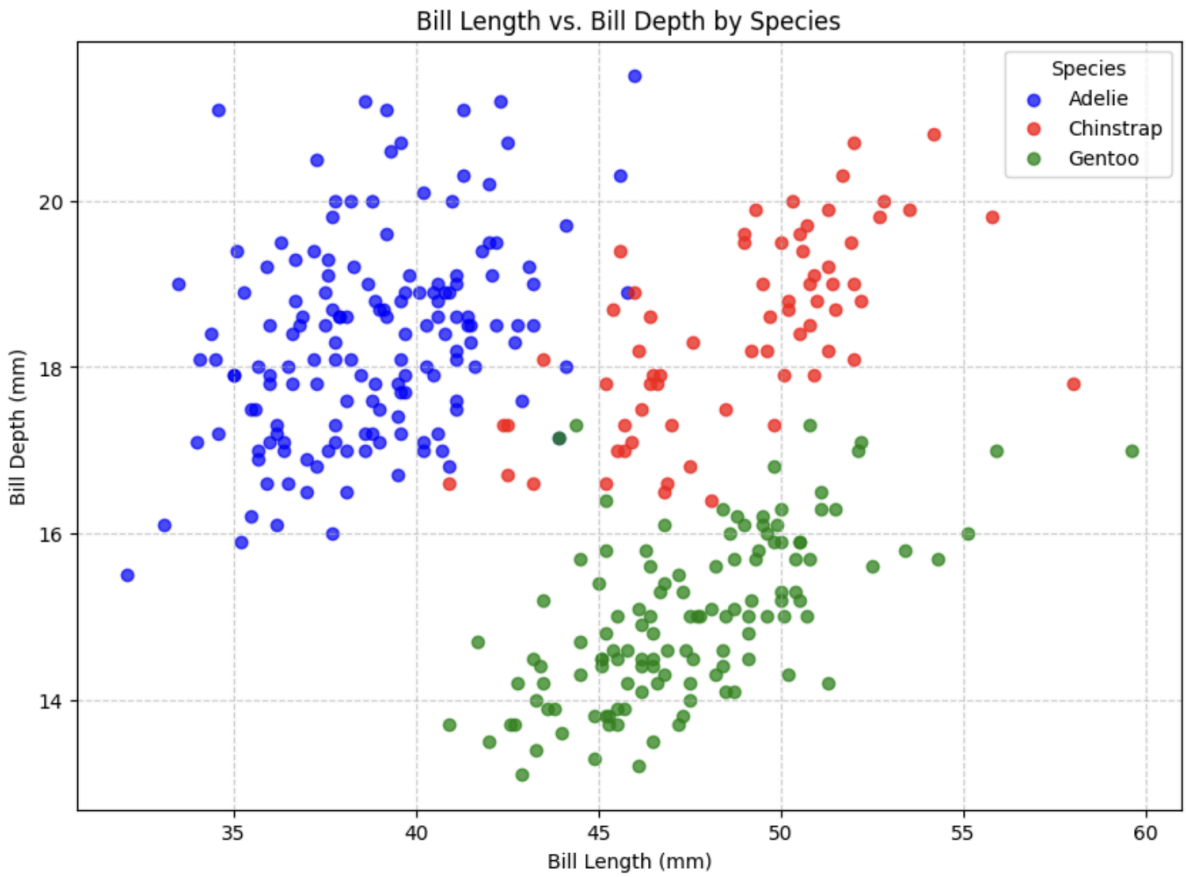


This bar chart shows the distribution of penguin species in the dataset. The three species represented are Adelie, Gentoo, and Chinstrap.

The visualization shows that Adelie penguins appear most frequently in the dataset, followed by Gentoo penguins, while Chinstrap penguins appear less frequently. This indicates that the dataset contains a larger number of Adelie observations compared to the other species.

Understanding the distribution of species helps provide context for the rest of the analysis and ensures that comparisons between species take into account the number of observations available for each group.

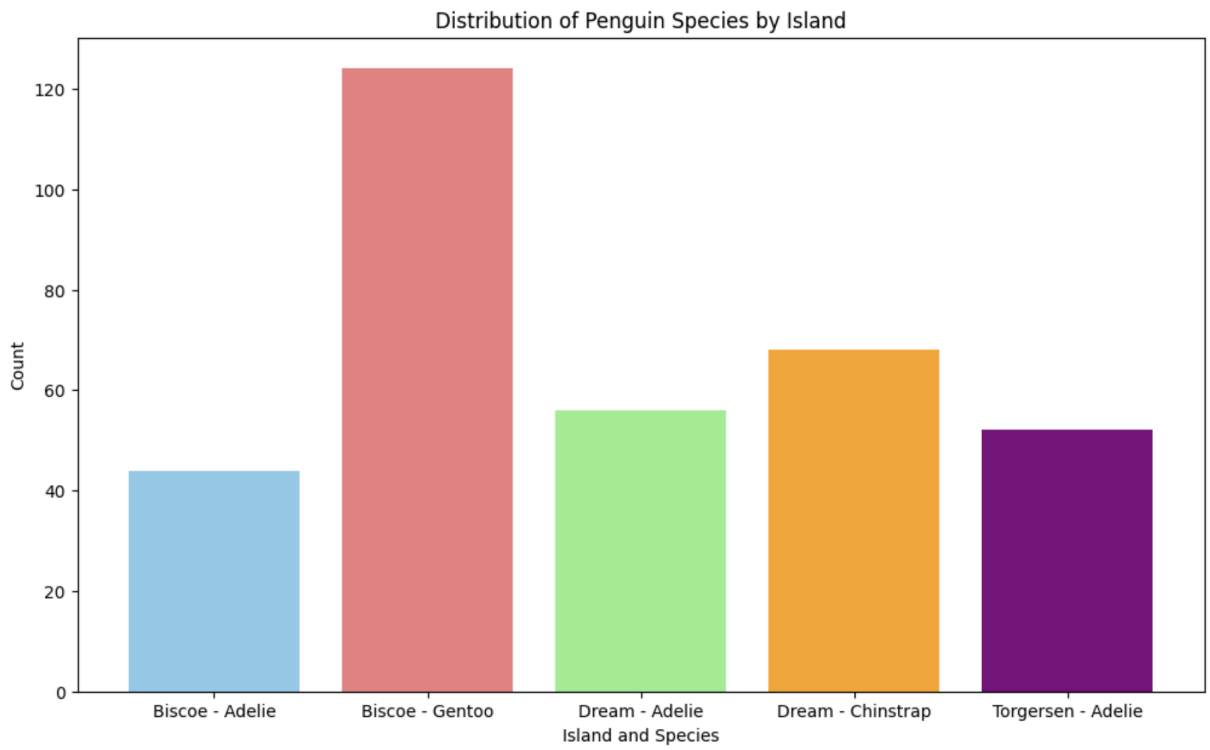
Figure 2 – Bill Length vs Bill Depth by Species



This scatter plot compares bill length and bill depth for the three penguin species. Each point represents a penguin observation, and different colors are used to distinguish between Adelie, Chinstrap, and Gentoo penguins.

The visualization shows that the species tend to form distinct clusters. Adelie penguins generally have shorter bill lengths and deeper bills, while Gentoo penguins tend to have longer bill lengths and shallower bill depths. Chinstrap penguins appear between these two groups. This pattern suggests that bill measurements can help differentiate penguin species.

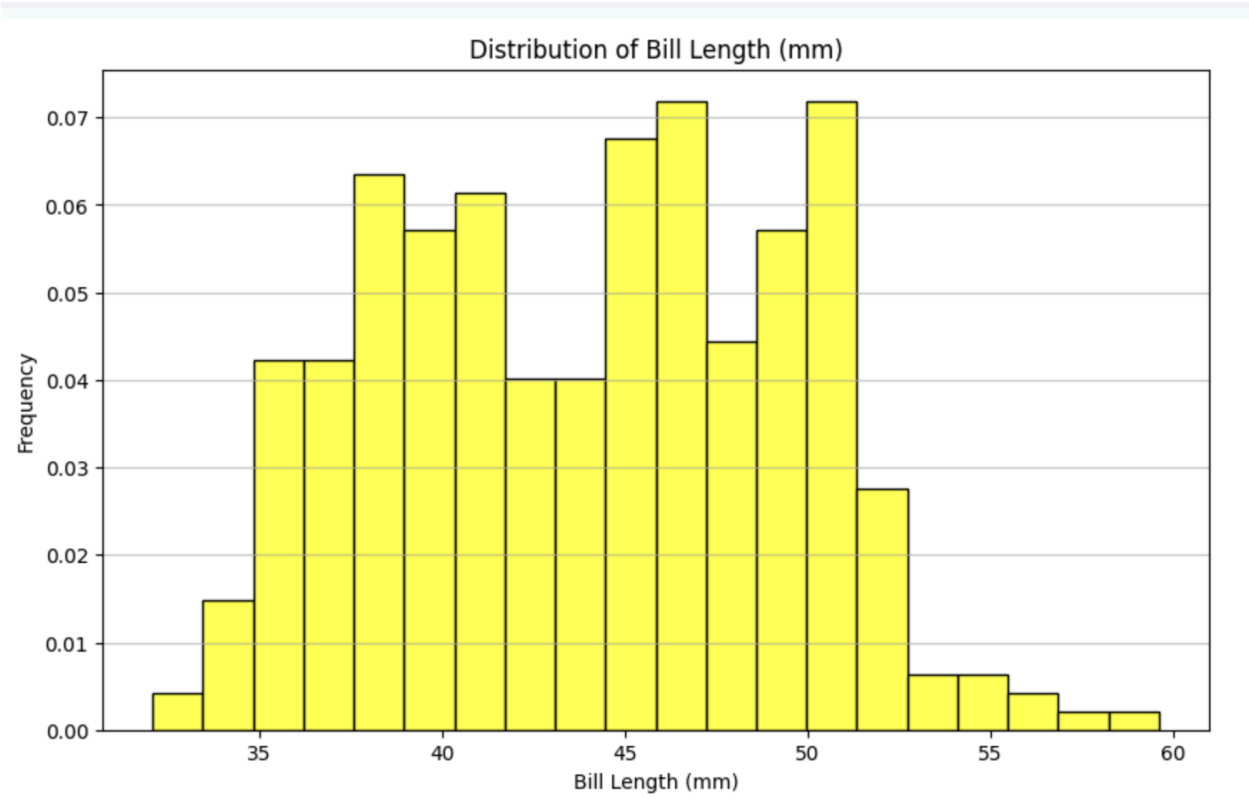
Figure 3 –Distribution of Penguin Species by Island



This histogram shows the distribution of bill length measurements for penguins in the dataset. The x-axis represents bill length in millimeters, while the y-axis represents the frequency of observations.

The visualization indicates that most penguins have bill lengths within a middle range, while fewer observations appear at the extreme low or high values. This distribution helps illustrate the typical range of bill length measurements among penguins in the dataset.

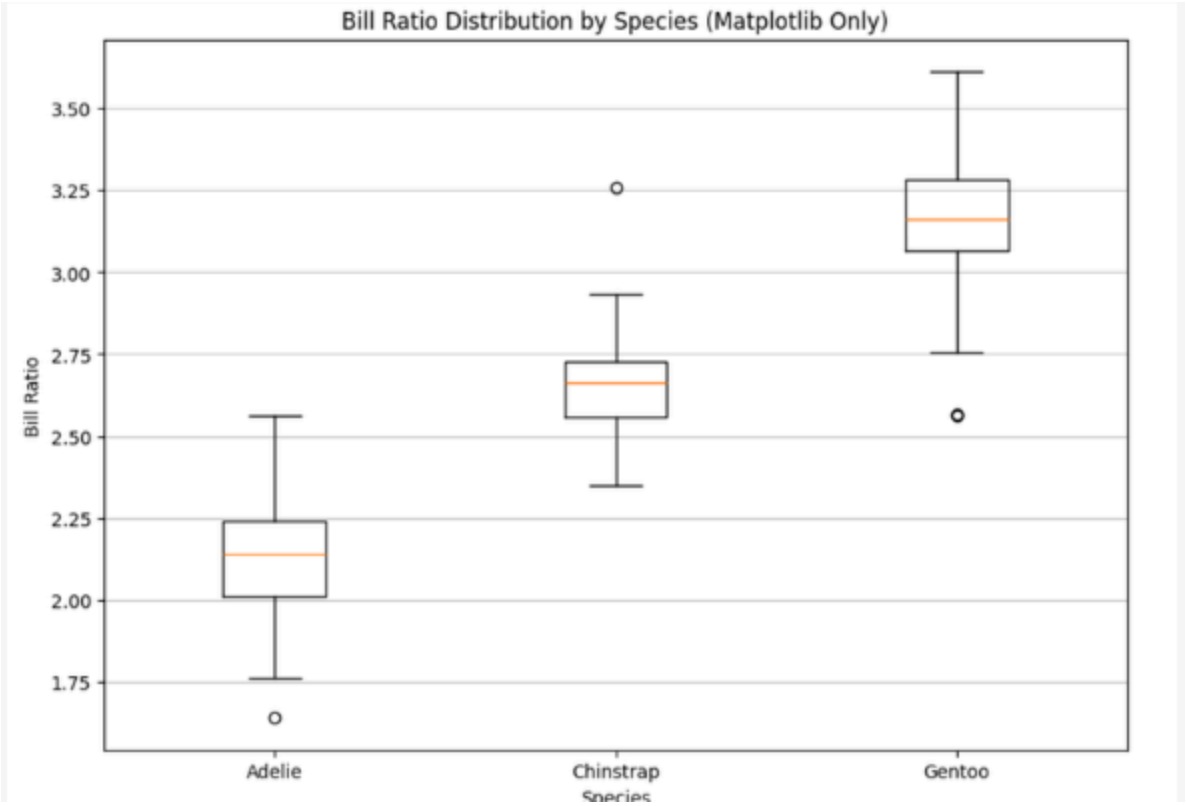
Figure 4 – Distribution of Bill Length



This bar chart shows how penguin species are distributed across different islands. The visualization combines island location and species to show where each type of penguin is commonly found.

The chart indicates that Gentoo penguins appear mainly on Biscoe Island, while Chinstrap penguins are mostly found on Dream Island. Adelie penguins appear across multiple islands, including Biscoe, Dream, and Torgersen. This suggests that the geographic location of islands influences where different penguin species are found.

Figure 5 – Bill Ratio Distribution by Species



This boxplot illustrates the distribution of the bill ratio (bill length divided by bill depth) for each penguin species in the dataset. The visualization highlights differences in the proportional relationship between bill length and bill depth across species. Gentoo penguins tend to have the highest bill ratio values, indicating relatively longer bills compared to their depth. Chinstrap penguins show intermediate values, while Adelie penguins generally have the lowest ratios. The boxplot also reveals the spread and variability of measurements within each species, providing insight into how bill proportions differ across penguin groups.

5. Key Insights

Several important insights were identified during the analysis of the penguin dataset.

First, different penguin species show clear differences in physical characteristics such as bill length, bill depth, flipper length, and body mass. These differences allow species to be visually distinguished when analyzing the data.

Second, the scatter plot of bill length versus bill depth shows that species tend to form distinct clusters. This suggests that bill measurements are strong indicators of species classification.

Third, the distribution of species across islands reveals that certain species are more common in specific locations. For example, Gentoo penguins are mainly found on Biscoe Island, while Chinstrap penguins are concentrated on Dream Island.

These insights demonstrate how exploratory data analysis and visualization can help reveal patterns and relationships within biological datasets.

6. Challenges Faced

During the project, our team faced two main challenges. First, as someone new to GitHub, we found the interface quite overwhelming at first, especially when trying to locate the collaboration settings to invite teammates. It took us some time to navigate through the repository settings to ensure everyone had access. Second, coordinating the workflow was a challenge because my active working hours were often later than the rest of the team. We were worried that our late-night updates might slow down the group's overall progress. However, by using GitHub as a central hub, we managed to bridge this time gap, allowing me to contribute effectively at my own pace without interrupting the team's momentum.

7. Learnings

This project completely changed our perspective on Data Science. Before this course, we thought analyzing data was an extremely difficult and solitary task. However, we learned that while the technical parts are complex, the process becomes much more manageable and even 'easier' when working in a team. By dividing the roles based on our strengths—like Eduardo handling the core code and me managing the repository—the workload felt less intimidating. we also learned that effective communication doesn't always require being the most talkative person; it’s about clear organization and ensuring the final deliverables, like the original CSV file and the repository, are properly maintained for the group